

Majority is not Enough: Bitcoin Mining is Vulnerable

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ANALYTICAL ABSTRACT

The paper asks whether the Bitcoin mining protocol rewards honest behaviour, as was widely assumed, or whether a minority coalition can profit by deviating. The authors describe Selfish-Mine, a strategy in which a pool withholds the blocks it finds, creates a private fork and publishes selectively so that honest miners waste effort on blocks that end outside the chain. They model the race between pool and honest miners as a Markov chain on the pool's lead, derive its stationary distribution and the pool's relative revenue as a function of its power α and of γ , the share of honest miners that follow the pool's block in a tie. The pool earns more than its fair share above a threshold of $(1 - \gamma)$ over $(3 - 2\gamma)$, which is one third at $\gamma = 0$ and vanishes at $\gamma = 1$. A protocol change that randomises tie-breaking raises the threshold to one quarter. This review rebuilds the algebra, confirms it by simulation, and argues that relative revenue, not profit, is what is established, and that the pool-growth and collapse claims are verbal rather than derived.

1 Introduction

1.1 Paper identification and classification

The paper is a theoretical study in the field of financial technology, and specifically in the economics of blockchain consensus. Its subfield is the incentive compatibility of proof-of-work mining. It is a model-based paper: a stylised Markov description of mining, an algorithm, closed-form expressions, and a simulation used as validation. It contains no empirical dataset beyond a few descriptive figures quoted in the introduction. The text read in full for this review is the arXiv version of the paper (arXiv:1311.0243, rendered as the latest version, which the arXiv listing dates 15 November 2013), including the abstract, the eight numbered sections, the conclusion, the reference list and Appendix 0.A. The published version, in the proceedings of the 2014 Financial Cryptography conference, was not read; differences between the two cannot be assessed here. The figures of the paper (the state machine, Figures 1 and 8, and the revenue and threshold plots, Figures 4 and 7) are not available in the text that was read, so statements about them rest on their captions and on the equations they plot; the review reproduces the plotted relations from the equations.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether a group of miners that controls less than half of the computing power can obtain a share of block rewards larger than its share of power by departing from the prescribed protocol.

WHY DOES THE QUESTION MATTER?

Bitcoin is a payment and settlement technology whose integrity depends on the incentives of the miners who maintain the shared ledger. If honest mining is not an equilibrium, then the usual security statement, that an attacker needs a majority of the computing power, is false, and the ledger can drift towards control by a single pool. For financial technology the question is about the foundations of a new class of settlement system: its value as a store of value or medium of exchange depends on decentralisation.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

The conventional view, represented in the paper by Barber et al. (2012), is that the protocol is incentive-compatible against minorities and that pool formation is benign, since a miner earns the same expected reward in a pool of any size. What was missing was a formal analysis of strategic withholding of blocks, and a quantification of the pool size above which it pays.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the Selfish-Mine strategy (Algorithm 1), analyses it with a state machine whose states are the pool's lead over the public chain, derives the stationary distribution and a closed form for pool revenue, validates it by a Monte Carlo simulator, and proposes a protocol modification. It finds a profitability threshold between zero and one third depending on γ , and a threshold of one quarter under the modification. The finding matters because it converts a qualitative fear into a number that can be compared with the observed concentration of mining pools.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The precise question is whether a pool with fraction $\alpha < 1/2$ of the total mining power can obtain a relative revenue (its expected share of blocks in the longest chain) larger than α by following a withholding strategy, given that a fraction γ of honest miners mine on the pool's block when two blocks compete, and, if so, how the minimum profitable size depends on γ .

2.2 Hypotheses

PAPER The claims can be stated as hypotheses.

- H1: honest mining is not an equilibrium, since for α above a threshold depending on γ the revenue of Selfish-Mine exceeds α .
- H2: the threshold is zero if the pool can win every tie ($\gamma = 1$), and at most one third otherwise.
- H3: above the threshold, revenue grows faster than proportionally with pool size, so rational miners join the largest selfish pool until it holds a majority.
- H4: random tie-breaking by honest miners ($\gamma = 1/2$) raises the threshold to one quarter and can be adopted progressively.

2.3 Existing literature and research gap

PAPER The authors cite Nakamoto (2008) for the protocol, Barber et al. (2012) for the view that the protocol is incentive compatible and for a scenario of single-entity control, Babaioff et al. (2012) for incentives to relay transactions, and Ron and Shamir (2013) for the transaction graph. They state that they are unaware of earlier work on the security of the blockchain itself. **CRITIQUE** The claim of novelty is plausible for the specific strategy, but Rosenfeld (2011) had already analysed reward systems and attacks for pools, so the attack surface of pooled mining was known in outline. The gap, formal analysis of block withholding by a minority pool, is real.

2.4 Claimed contribution

The contribution is the strategy, its analysis, a pool-formation argument, and a fix. The first two are demonstrated mathematically within the model; the third is an informal argument; the fourth is argued but not analysed under partial adoption or against alternative attacks.

3 Theoretical Framework and Mechanism

3.1 The model

PAPER There are n miners with powers m_i summing to one. Miner i finds a block on the head it mines after an exponentially distributed time with mean m_i^{-1} . Honest miners follow the protocol (mine on the longest chain, the first one they hear of in a tie). One pool, of power α , follows a strategy; the rest, of power $1 - \alpha$, are honest, which is without loss of generality since honest miners may be a single group or many. A pool's relative revenue is the expected fraction of blocks of the longest chain that it mined. Because block discovery is memoryless, the next block found is the pool's with probability α and honest with probability $1 - \alpha$, independently of the past; the dynamics are therefore a Markov chain driven by Bernoulli events of probabilities α and $1 - \alpha$. Propagation delay is taken as negligible except for ties, where γ is the share of honest miners that adopt the pool's block.

3.2 The Selfish-Mine strategy

PAPER The strategy depends only on the lead $\Delta = \text{length}(\text{private chain}) - \text{length}(\text{public chain})$ and is driven by two kinds of events. Table 1 restates Algorithm 1 as a table of cases.

Table 1. Selfish-Mine (Algorithm 1) restated as a table of cases

Event	Lead before the event	Action of the pool
Pool finds a block	Any lead except a tie with a branch of length one	Append to the private chain; the lead rises by one
Pool finds a block	Tie of two branches of length one	Publish the two-block private chain; the pool earns two blocks
Others find a block	0 (no private branch)	Adopt the public chain; the others earn one block
Others find a block	1	Publish the single private block, creating a tie of two branches of length one
Others find a block	2	Publish the whole private chain; the pool earns two blocks and the lead returns to zero
Others find a block	More than 2	Publish the first unpublished private block; the lead falls by one

Source: restated by the reviewer from Algorithm 1 and Section 3.2 of the paper (official content, reorganised).

INFERENCE The economic logic is the following. A block found by the pool is kept secret if the pool is ahead because its eventual publication can invalidate honest blocks; the honest miners continue to mine on a shorter chain and, when the pool reveals its longer branch, their work is discarded. The pool incurs risk only in the state of lead one, where a single honest block creates a tie that is resolved by γ . Selfish mining therefore works by transferring orphaned blocks from the pool to the honest miners: the total rate of blocks in the longest chain falls, and because the difficulty adjusts to keep that rate at one block per ten minutes, the pool's fraction of the remaining blocks rises.

3.3 State probabilities

PAPER The states are the lead $0, 1, 2, \dots$ with an extra state $0'$ for a tie of two public branches of length one. Each pool block has frequency α and each honest block frequency $1 - \alpha$. Balance across cuts of the state graph (Appendix 0.A) yields

$$\begin{aligned} \alpha p_0 &= (1 - \alpha)(p_1 + p_2), & p'_0 &= (1 - \alpha)p_1, & \alpha p_1 &= (1 - \alpha)p_2, \\ \alpha p_k &= (1 - \alpha)p_{k+1} \quad (k \geq 2), & \sum_{k \geq 0} p_k + p'_0 &= 1. \end{aligned} \quad (1)$$

p_k	Stationary probability that the pool's lead is k blocks ($k = 0$: no branch).
p'_0	Stationary probability of a tie between two public branches of length one.
α	Pool share of mining power, which is the probability that the next block is the pool's.

The equations are the statement that the flow of probability up across each cut equals the flow down. The solution, which the paper gives without the algebra, runs as follows. The third and fourth equations give a geometric tail, $p_k = \left(\frac{\alpha}{1-\alpha}\right)^{k-1} p_1$ for $k \geq 1$. Substituting $p_2 = \frac{\alpha}{1-\alpha} p_1$ in the first equation gives $\alpha p_0 = (1 - \alpha)p_1 + \alpha p_1 = p_1$, so $p_0 = p_1/\alpha$. Writing $q = \alpha/(1 - \alpha) < 1$ (this is where $\alpha < 1/2$ is needed), the geometric sum is $\sum_{k \geq 1} q^{k-1} = (1 - \alpha)/(1 - 2\alpha)$, and normalisation reads

$$\begin{aligned} 1 &= p_1 \left[\frac{1}{\alpha} + \frac{1 - \alpha}{1 - 2\alpha} + (1 - \alpha) \right] \\ &= p_1 \frac{(1 - 2\alpha) + \alpha(1 - \alpha) + \alpha(1 - \alpha)(1 - 2\alpha)}{\alpha(1 - 2\alpha)} = p_1 \frac{1 - 4\alpha^2 + 2\alpha^3}{\alpha(1 - 2\alpha)}, \end{aligned} \quad (2)$$

which gives the paper's equations (2) to (5):

$$p_1 = \frac{\alpha - 2\alpha^2}{2\alpha^3 - 4\alpha^2 + 1}, \quad p_0 = \frac{p_1}{\alpha}, \quad p'_0 = (1 - \alpha)p_1, \quad p_k = \left(\frac{\alpha}{1 - \alpha}\right)^{k-1} p_1 \quad (k \geq 2). \quad (3)$$

The stationary distribution exists only for $\alpha < 1/2$: at $\alpha = 1/2$ the ratio q reaches one, the lead is a driftless random walk and no stationary law exists.

3.4 Revenue by event

PAPER The reward of a block is earned only if it ends in the longest chain, so the authors list the cases (a) to (h) in which a block is credited. Their frequencies and rewards give, per unit of mining events,

$$r_{\text{others}} = p'_0 \gamma (1 - \alpha) \cdot 1 + p'_0 (1 - \gamma) (1 - \alpha) \cdot 2 + p_0 (1 - \alpha) \cdot 1, \quad (4)$$

$$r_{\text{pool}} = p'_0 \alpha \cdot 2 + p'_0 \gamma (1 - \alpha) \cdot 1 + p_2 (1 - \alpha) \cdot 2 + P[i > 2] (1 - \alpha) \cdot 1, \quad (5)$$

γ Share of honest miners that mine on the pool's block in a tie (the rest mine on the honest block).

$P[i > 2]$ Probability that the lead exceeds two, equal to $\sum_{k \geq 3} p_k = p_1 \alpha^2 / ((1 - \alpha)(1 - 2\alpha))$.

$r_{\text{pool}}, r_{\text{others}}$ Expected blocks credited to the pool and to the others per mining event.

Each term has an interpretation. In a tie the pool wins two blocks if it finds the next block ($\alpha \cdot 2$); if honest miners find it on the pool's branch ($\gamma(1 - \alpha)$) the two sides split one block each; if they find it on their own branch ($(1 - \gamma)(1 - \alpha)$) the honest side wins two. With no private branch an honest block simply earns one. From a lead of two an honest block triggers publication of the whole private chain, which earns the pool two blocks, and from a lead above two each honest block is paid back by the pool revealing one of its own, which earns the pool one. The sum $r_{\text{pool}} + r_{\text{others}}$ is below one: the missing mass is the orphan rate. The pool's relative revenue is

$$R_{\text{pool}} = \frac{r_{\text{pool}}}{r_{\text{pool}} + r_{\text{others}}} = \frac{\alpha(1 - \alpha)^2 (4\alpha + \gamma(1 - 2\alpha)) - \alpha^3}{1 - \alpha(1 + (2 - \alpha)\alpha)}. \quad (6)$$

The paper writes this with an ellipsis between the two expressions; the reviewer verified numerically, with the equations (3) to (5), that the ratio equals Equation (6) to machine precision at several values of (α, γ) (for instance $\alpha = 0.25, \gamma = 0.5$ gives exactly 0.25).

3.5 The profitability threshold

PAPER The pool gains if $R_{\text{pool}} > \alpha$. The paper states the result (Observation 1) without algebra; the derivation is short. Since the denominator is $D = 1 - \alpha - 2\alpha^2 + \alpha^3 > 0$ for $\alpha < 1/2$, dividing Equation (6) by α and cross-multiplying gives the condition $(1 - \alpha)^2 (4\alpha + \gamma(1 - 2\alpha)) - \alpha^2 > D$, that is

$$\gamma(1 - 2\alpha)(1 - \alpha)^2 > 1 - 5\alpha + 7\alpha^2 - 3\alpha^3 = (1 - \alpha)^2(1 - 3\alpha) \iff \alpha > \frac{1 - \gamma}{3 - 2\gamma}. \quad (7)$$

The threshold is $1/3$ at $\gamma = 0$, $1/4$ at $\gamma = 1/2$ and 0 at $\gamma = 1$. The slope $dR_{\text{pool}}/d\alpha$ exceeds one above the threshold (Observation 2): between $\alpha = 0.25$ and 0.30 at $\gamma = 0.5$ the reviewer's calculation from Equation (6) gives a slope of about 1.5, so revenue per unit of power rises with pool size and larger pools are more attractive. At $\alpha \rightarrow 1/2$ Equation (6) gives $R_{\text{pool}} = 1$ for any γ , as expected: a pool with half of the power eventually controls the whole chain.

3.6 The protocol modification

PAPER The current rule, to mine on the first branch heard, lets a pool with well-connected sybil nodes (zero-power nodes that relay blocks) win ties, pushing γ towards one. The proposed rule is that a miner who learns of competing branches of equal length propagates all of them and chooses uniformly at random which to mine on, which yields $\gamma = 1/2$ and a threshold of $1/4$ by Equation (7). **INFERENCE** If only a fraction f of honest miners adopt it and the pool wins ties with the others, then $\gamma = (1 - f) + f/2 = 1 - f/2$ and the threshold is $\frac{f/2}{1+f}$, which rises from 0 at $f = 0$ to $1/4$ at $f = 1$; the paper's statement that adoption is progressive is therefore correct in direction, but the gain is concave in f and is not proportional to it.

4 Data and Computational Strategy

4.1 Data and calibration

PAPER There is no dataset. The introduction quotes a hash rate above 42×10^{18} operations per second and a market capitalisation of about 1.5 billion dollars in October 2013 (from public chart sites cited in the paper), the rate of accidental forks (about once per 60 blocks, from Decker and Wattenhofer 2013), and observation that some pools already exceeded 25 percent of the power (a blog post cited as a source). **CRITIQUE** The empirical support for the preconditions is thin: none of these is a dataset on which the model is calibrated.

4.2 Simulation

PAPER A simulator with 1000 miners of equal power, of which 1000α form a pool running Algorithm 1, replaces the cryptographic puzzle by Monte Carlo block discovery; block propagation is instantaneous except for ties, which are split so that a fraction γ of the others mine on the pool's branch. The paper reports that the simulation matches Equation (6) in Figure 4. The number of runs and the confidence intervals are not stated in the text. **EVIDENCE** As an independent check, the reviewer implemented the algorithm as an event-by-event chain (Bernoulli events with probabilities α and $1 - \alpha$ and the action table above) and ran 300,000 events per point. At $(\alpha, \gamma) = (0.25, 0.5)$ the simulated relative revenue is 0.2500 against 0.2500; at $(0.40, 0)$ it is 0.4820 against 0.4837; at $(0.30, 1)$ it is 0.3823 against 0.3806; at $(0.10, 0.5)$ it is 0.0735 against 0.0724. The agreement confirms the algebra from the state machine to the closed form; it does not validate the state machine as a description of the Bitcoin network, because the reviewer's chain and the paper's simulator share the same modelling assumptions.

4.3 Where does the identifying variation come from?

CRITIQUE There is no identification in the econometric sense. The results are exact consequences of the assumptions: independent exponential block times, a single deviating pool, instantaneous propagation, a fixed γ , and revenue measured as a share of the longest chain. The relevant question is therefore which assumptions drive the result, taken up in the critique.

4.4 Limitations of the setting

- External validity: the model has one deviating pool and treats honest miners as a block. Several strategic pools, entry and exit, and heterogeneous propagation are outside it.
- Measurement of γ : the parameter is exogenous and is never measured in the paper; the claim that it is close to one in the actual protocol rests on an argument about sybil nodes, not on data.
- Costs and risk: the model has no operating costs, no variance of revenue, no discounting and no price of the currency.
- Look-ahead and selection concerns do not apply because there is no sample.

5 Results

5.1 Pool revenue as a function of power

EVIDENCE Table 2 evaluates Equation (6) on a grid, together with the share of blocks that remain in the chain.

Table 2. Relative revenue of a Selfish-Mine pool, from Equation (6)

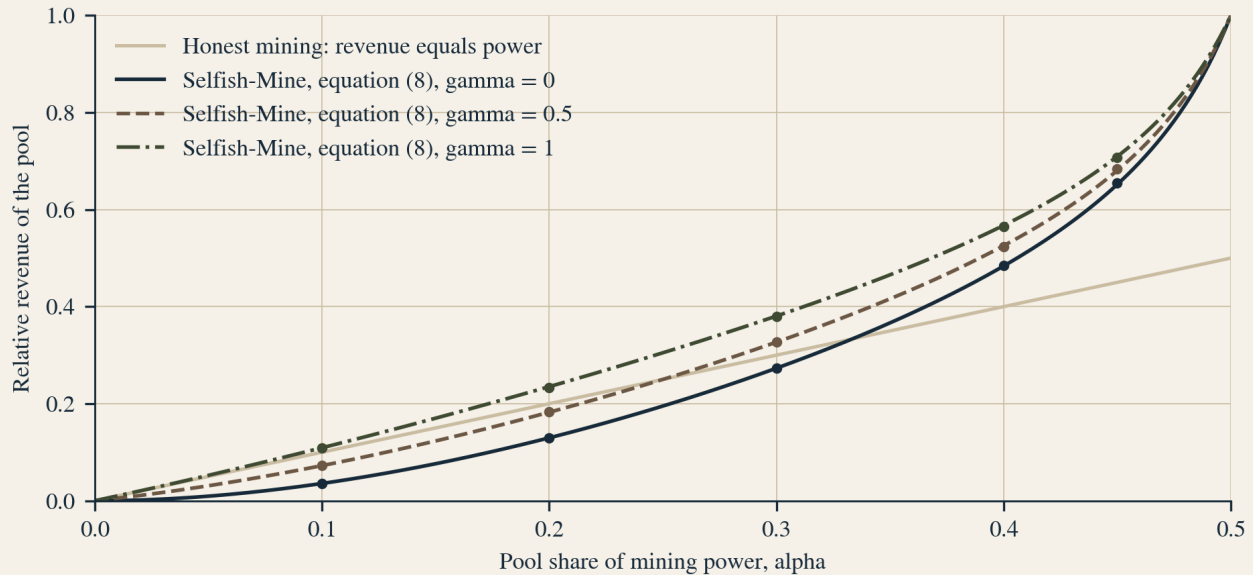
Pool share α	$R_{\text{pool}}, \gamma = 0$	$R_{\text{pool}}, \gamma = 0.5$	$R_{\text{pool}}, \gamma = 1$	Blocks kept in the chain
0.10	0.036	0.072	0.109	0.916
0.20	0.130	0.182	0.235	0.850
0.25	0.195	0.250	0.305	0.820
0.30	0.273	0.327	0.381	0.788
0.33	0.327	0.379	0.430	0.767
0.40	0.484	0.526	0.567	0.705
0.45	0.652	0.681	0.709	0.634

Source: reviewer's calculation from Equations (3) to (6) (generated, not copied from the paper). The last column is $r_{\text{pool}} + r_{\text{others}}$, which does not depend on γ .

EVIDENCE Interpretation: the table answers how much the pool gains. The key quantity is the comparison of each entry with α . With $\gamma = 1/2$ a pool with 30 percent of the power earns 32.7 percent of the rewards, an excess of about 9 percent of its fair share; at 40 percent it earns 52.6 percent, a gain of 31 percent. With $\gamma = 0$, even a pool of 33 percent does not yet break even (32.7 percent of the rewards against 33 percent of the power). The table also shows what the pool pays for the gain:

when the pool has 30 percent, only 78.8 percent of mined blocks remain in the chain. The table fails to show anything about profit, since costs and time are absent.

Figure 1. Relative revenue of a Selfish-Mine pool against its share of mining power, for three values of the tie parameter

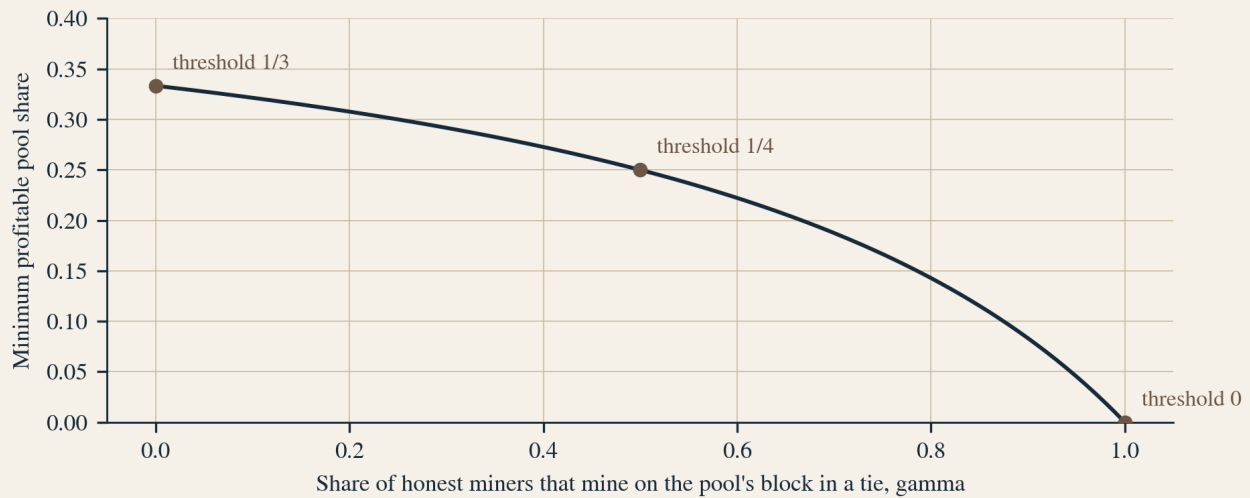


Source: lines from Equation (8) of the paper (official formula, plotted by the reviewer); points are the reviewer's own Monte Carlo of the Selfish-Mine chain (generated), 300,000 events per point. The grey line is honest mining.

EVIDENCE Figure 1 shows that all three curves coincide at zero and at one half, are convex above the honest line, and fan out with γ : the gain from raising γ is largest for mid-sized pools. The relation is a model result, not a causal estimate. The simulated points lie on the curves, except for small deviations at the larger pool sizes, where the lead process has long excursions and the Monte Carlo error is larger.

5.2 The threshold

Figure 2. Minimum profitable pool share as a function of the tie parameter gamma



Source: Equation (9) of the paper (official formula, plotted by the reviewer). No data are involved.

EVIDENCE Figure 2 reproduces the paper's Figure 7: the threshold falls from one third to zero as γ rises, and the current protocol, in the paper's argument, sits at the right-hand end. The curve is mildly convex: most of the reduction in the threshold from the fix (from $\gamma = 1$ to $\gamma = 1/2$) is bought by the first half of the move.

5.3 Pool formation

PAPER Section 5 argues that miners will join a selfish pool above the threshold because revenue per member increases with pool size, that pool operators will accept new members for the same reason, and that nothing stops the pool from reaching a majority, at which point the currency is no longer decentralised. **CRITIQUE** This argument has no model: there is no miner-level choice problem, no cost of switching pools, no response by other pools and no equilibrium concept. It is plausible but is a conjecture supported by the monotonicity of R_{pool} in α .

6 Robustness and Additional Tests

CRITIQUE The paper has few robustness exercises. Table 3 states each concern, the test reported (or its absence), and the reviewer’s assessment.

Table 3. Robustness tests: concern, test, result and assessment

Concern	Test reported	Result	Does it resolve the concern?
Algebraic error in the closed form	Monte Carlo simulator with 1000 miners	Matches Equation (8), according to Figure 4	Yes for the algebra (confirmed independently by the reviewer); not for realism
Dependence on the tie parameter	Curves for several γ ; threshold as a function of γ	Threshold from 0 to 1/3	Yes within the model; γ is not measured
Block propagation delay	Assumed negligible, except in ties	None	No: delays and the orphan rate of honest miners are ignored
Optimality of the strategy	None	Not addressed	No: other strategies could do better
Detection and counter-attack	Verbal discussion in Section 8	Argued to be ineffective	No: no model of detection
Altruistic miners	Verbal discussion in Section 8	Conjectured to be too few	No: assertion
Effect of the fix	Verbal and algebraic ($\gamma = 1/2$)	Threshold 1/4	Partly: partial adoption and sybil adaptation are not analysed

Source: reviewer’s assessment of Sections 4 to 8 of the paper.

CRITIQUE The tests address the algebra, not the model. There is no placebo equivalent (a theory paper does not need one) but there is no sensitivity analysis to the assumptions that matter: propagation speed, number of pools, costs, retargeting and the price of the currency.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Within the stated model, honest mining is not an equilibrium against a single deviating pool: the relative revenue of Selfish-Mine is given exactly by Equation (6), exceeds the pool’s share of power above the threshold $(1-\gamma)/(3-2\gamma)$, and the threshold is at most one third. The algebra from the state machine to the closed form is correct (verified by the reviewer analytically and by simulation).

SUPPORTED BUT CONDITIONAL

That the current protocol has a threshold close to zero depends on the assumption that a pool can drive γ towards one, through sybil relay nodes. That a random tie-breaking rule raises the threshold to one quarter depends on all honest miners adopting it and on the pool being unable to bias which block honest miners receive first.

SUGGESTIVE

That rational miners will join the selfish pool and that the pool will grow until it holds a majority, and that Bitcoin would then collapse. The monotonicity of revenue in pool size makes it plausible; there is no model of the choices of miners or of the response of the price.

NOT ESTABLISHED

That selfish mining is profitable once costs, timing and risk are counted; that Selfish-Mine is the best deviation; that its threshold is a threshold for the real network; that detection is impossible; that altruistic miners are too few to matter; that γ in the live protocol is close to one.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- A concrete algorithm and a closed-form analysis, which let anyone compute the revenue of the strategy for any α and γ and compare it with the observed concentration of pools.
- A mitigation that is backward-compatible and progressive, which does not require a hard fork and so has a realistic path to adoption.
- A clear threshold result, including the sharp statement that no majority assumption protects against the strategy.

STRENGTHS IDENTIFIED INDEPENDENTLY

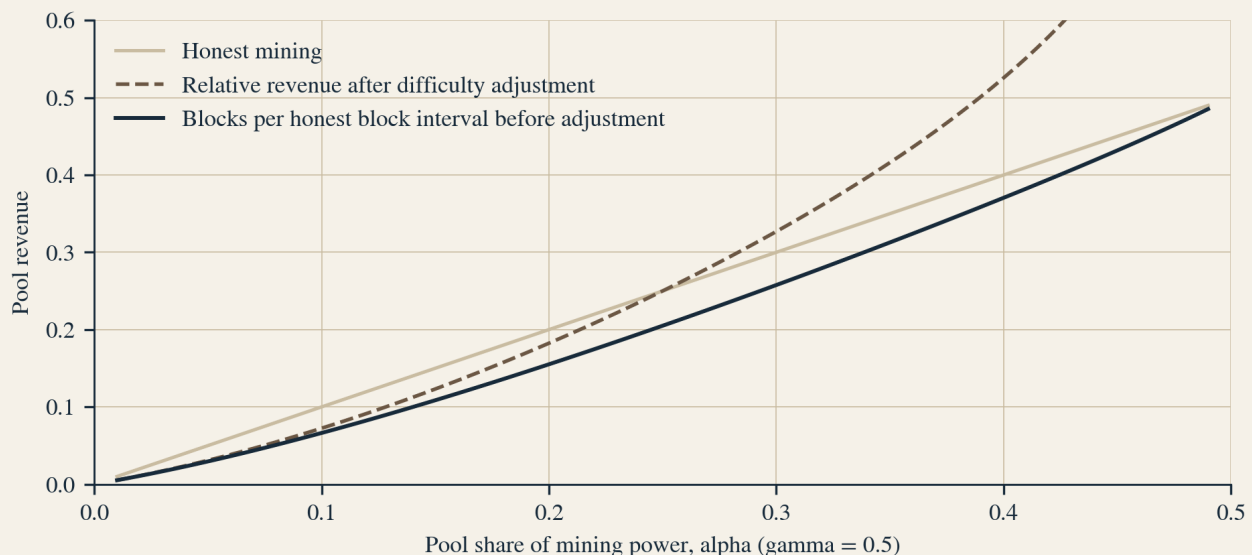
- Parsimony: one parameter beyond power, γ , captures the whole role of the network, which makes the model transparent and easily extended, as the later literature shows.
- The result is exact in the model and robust to the details of honest miners: it does not matter whether they are one agent or many.
- The revenue decomposition by event (cases (a) to (h)) is verifiable line by line; the reviewer was able to reproduce it from the text.
- A genuinely surprising negative result for a protocol assumed secure: it shows that incentive compatibility must be checked and not assumed.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

CRITIQUE Problem. The criterion is relative revenue, the pool's share of blocks in the longest chain, not profit per unit of time. The paper argues that the difficulty adjusts to keep the chain rate at one block per ten minutes, but the adjustment is not instantaneous (the Bitcoin protocol retargets every 2016 blocks, a parameter that the paper does not state). **Why it matters.** Before the adjustment the pool's blocks per unit of time fall below its honest rate: in the model, the pool earns r_{pool} blocks per honest block interval, and Figure 3 shows that $r_{\text{pool}} < \alpha$ for every $\gamma < 1$ and every $\alpha < 1/2$. At $\alpha = 0.30$ and $\gamma = 0.5$ the pool earns 0.258 blocks per interval against 0.300 when honest, a loss of 14 percent, until the difficulty falls. **Potential bias.** The threshold of Equation (7) is a threshold for long-run relative revenue and is too low as a threshold for profit over any finite horizon, since the pool must first bear a transition cost. **How it could be tested.** Compute the expected profit over a finite horizon with the retargeting rule, discounting and operating cost (Idea 1 below).

Figure 3. Pool revenue before and after the difficulty adjustment, for gamma equal to 0.5



Source: reviewer's calculation from Equations (6) to (8) of the paper (generated). The line before the adjustment is r_{pool} , expressed in blocks per honest block interval; the line after is R_{pool} . The grey line is honest mining.

INFERENCE A numerical illustration, under assumptions labelled as the reviewer's (144 blocks per day at the honest rate, retargeting every 2016 blocks, no cost, no change of the block subsidy): at $\alpha = 0.30$ and $\gamma = 0.5$, the total chain rate is 0.788 so the first window lasts $2016/0.788$ blocks intervals, about 17.8 days instead of 14. In that period the pool earns about 37.1 blocks per day against 43.2 as an honest miner, a deficit of about 109 blocks. After the adjustment it earns $0.3269 \times 144 = 47.1$ blocks per day, a gain of 3.9 blocks per day, so the deficit is recovered after about 28 further days. The strategy is therefore profitable only for pools that are patient and not closely watched, and the horizon is short in cryptocurrency terms; but the correct statement is a threshold in time as well as in size.

OTHER WEAKNESSES

- **Optimality. Problem:** Selfish-Mine is one strategy, hard-coded in Algorithm 1. **Why it matters:** the threshold of the best strategy is lower; later work on optimal strategies (Sapirshtein, Sompolinsky and Zohar 2016) reports thresholds below one third even at $\gamma = 0$, which the reviewer recalls to be about 23 percent but has not verified in this review. **Potential bias:** the paper’s one-third bound is not a bound on all attacks, and its statement that the protocol is never safe above one third is right but its safety of lower pools is not. **Test:** a Markov decision process over all withholding policies.
- **Exogenous γ . Problem:** γ is a constant, though it results from network topology and from the pool’s connectivity. **Why it matters:** the threshold is highly sensitive to it (from 0.33 to 0). **Test:** a network propagation model with latency and sybil nodes, calibrated on observed fork resolution.
- **Price endogeneity. Problem:** the value of a block reward depends on the market price of Bitcoin, which would respond to an attack or to a collapse. **Why it matters:** a rational attacker with a stake in the currency gains less than the model suggests. **Potential bias:** overstated incentives. **Test:** a model with a price process and an event study around known thresholds.
- **Static, single deviation. Problem:** honest pools do not respond, and the entry of several selfish pools is not analysed. **Why it matters:** two selfish pools hurt each other, and a response by honest pools (leaving the largest pool) can keep the system below the threshold. **Test:** a multi-player game.
- **Unsupported claims in Section 8. Problem:** detection, altruism and the cost of counter-attacks are asserted. **Why it matters:** these claims carry the policy conclusion that the danger is imminent. **Test:** empirical analysis of fork and orphan statistics.
- **A cross-reference inconsistency. Problem:** Section 5 cites a Theorem 1 that is not stated in the text read, where the result is stated as Observation 1. **Why it matters:** minor, but it suggests that the proof structure of the version read was reorganised.

8 Contradictions and Edge Cases

CRITIQUE The questions below try to break the model, using direct reasoning and calculation on the equations of the paper.

8.1 Where the mechanism should fail

- **Large pools.** The state probabilities require $\alpha < 1/2$. As $\alpha \rightarrow 1/2$ the stationary distribution degenerates (the lead has no drift) and Equation (6) tends to one for every γ . For $\alpha \geq 1/2$ the pool can ignore the honest chain entirely, and the strategy is unnecessary, as the paper notes.
- **Very small pools.** For $\gamma < 1$, $R_{\text{pool}} \approx \gamma\alpha < \alpha$ as $\alpha \rightarrow 0$: the strategy loses. At $\gamma = 1$ the result relies on a second-order effect, which is below the noise of any realistic sample (Table 2 gives a gain of 9 percent of α at $\alpha = 0.1, \gamma = 1$).

- Slow propagation. If honest miners hear of a block seconds later than the pool, then honest blocks themselves fork with probability near one in 60 blocks (paper’s own figure), and honest forks give the pool further chances to publish; the paper ignores this baseline and the cases where it interacts with selfish mining.
- Costs. If the cost of mining is proportional to power, then profit is $R_{\text{pool}} \times (\text{reward}) - \alpha \times c$ and any positive operating margin shrinks the threshold’s gain. A pool that is barely above the threshold earns almost nothing for the risk it takes.

8.2 Alternative mechanisms that generate similar outcomes

- Propagation advantage without withholding: a well-connected honest pool earns a share above α because its blocks are heard first, which is observationally close to the effect that $\gamma > 1/2$ gives. Distinguishing selfish mining from a connectivity advantage is a statistical problem that the paper does not address.
- Block withholding inside pools, mentioned in Section 7 as unrelated, interacts with Selfish-Mine: a pool member who withholds changes the pool’s effective power.

8.3 Effect of the proposed fix, with partial adoption

INFERENCE With a fraction f of honest miners adopting random tie-breaking and the pool controlling the propagation to the others, $\gamma = 1 - f/2$ and the threshold is $\frac{f/2}{1+f}$ (Section 3.6). At $f = 0.5$ the threshold is about 0.167, at $f = 0.8$ it is about 0.22, and it reaches 0.25 only at full adoption. Note that the fix does not touch the effect of selfish mining above one quarter, and does not change the revenue curve above the threshold, only the threshold.

9 Further Literature

INFERENCE Table 4 places the paper in the literature. The paper cites few antecedents; most of the connections below are later developments.

Table 4. Literature connections

Paper	Shared idea	Relationship	Why it matters
Nakamoto (2008)	Proof-of-work chain secure against an attacker with a minority of the power	Foundation; the claim that Selfish-Mine refutes in its incentive form	Defines the protocol and the honest strategy
Rosenfeld (2011)	Analysis of pool reward schemes and attacks on pools	Preceded; same economics of pools	Shows that pool design matters for incentives
Barber et al. (2012)	Incentive compatibility as a property of Bitcoin	Cited as the conventional view that the paper refutes	Frames the debate

Paper	Shared idea	Relationship	Why it matters
Eyal (2015)	Game between pools that attack each other by infiltration	Extension by the same first author	Shows the equilibrium among several pools
Sapirshtein, Sompolinsky and Zohar (2016)	Optimal selfish mining by dynamic programming	Improves the strategy and lowers thresholds	Tests whether Selfish-Mine is optimal
Nayak et al. (2016)	Stubborn mining and eclipse attacks	Extends the strategy space	Shows additional gains from other deviations
Kiayias et al. (2016)	Mining as a game with several players	Formalises the strategic game	Provides the equilibrium analysis that the paper omits
Garay, Kiayias and Leonardos (2015)	Formal security of the chain protocol	Provides a security model for the honest majority case	Contrasts security of the protocol with incentive of miners
Heilman (2014); Zhang and Preneel (2017)	Defences against selfish mining	Alternative to the paper's tie-breaking rule	Compare mitigations and thresholds
Grunspan and Pérez-Marco (2018)	Profitability of selfish mining over time with difficulty retargeting	Contradicts or qualifies the relative-revenue criterion	Addresses the main weakness identified here
Budish (2018)	Economic limits of proof-of-work security	Complementary: cost of attack versus value at stake	Links the technical vulnerability to economic value
Biais et al. (2019)	Equilibria of the blockchain game, forks and multiple equilibria	Finance-theoretic counterpart	Interprets forks as equilibrium phenomena
Easley, O'Hara and Basu (2019)	Fees and miner behaviour	Applies microstructure to mining	Connects mining incentives with fee markets

Source: reviewer's compilation. References are listed in full in Section 14. The relationships are characterised from the contribution of each paper as the reviewer understands it, and not from the paper reviewed; those papers were not read in full for this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** profit versus relative revenue. *Why it matters:* an attack is chosen on profit. *Existing limitation:* the criterion ignores retargeting, cost and discounting. *Way forward:* finite-horizon dynamic programming with the retargeting rule.
- **Gap:** measurement of γ . *Why it matters:* it drives the threshold from 0 to one third. *Existing limitation:* it is a free parameter. *Way forward:* estimate it from block propagation data and observed tie resolution.
- **Gap:** the pool-growth dynamics. *Why it matters:* the collapse conclusion depends on it. *Existing limitation:* verbal argument. *Way forward:* a model of pool choice with switching costs.

10.2 Literature-level gaps

- **Gap:** evidence of selfish mining in actual blocks. *Why it matters:* a theory of attack is incomplete without evidence of its use. *Existing limitation:* withholding is hard to observe. *Way forward:* statistical detection from timestamps, fork lengths and orphan rates.
- **Gap:** link between consensus incentives and asset valuation. *Why it matters:* the cost to attackers and the value of the asset are endogenous to each other. *Way forward:* joint models of mining and price.

10.3 Methodological gaps

- **Gap:** the model is a stationary Markov chain with no time dimension. *Way forward:* continuous-time models with retargeting and stopping times.
- **Gap:** the simulation is described without error bars. *Way forward:* report replications and confidence intervals, and open code.

11 Research Opportunities

EXTENSION The ideas below follow from the weaknesses identified. Feasibility and data statements are the reviewer's estimates.

Idea 1. Finite-horizon profitability of selfish mining with difficulty retargeting

RESEARCH QUESTION. For which pool sizes and horizons does Selfish-Mine have a positive expected net present value once the 2016-block retargeting rule, discounting and operating cost are included?

MOTIVATION. Relative revenue overstates profitability during the adjustment window.

HYPOTHESIS. The break-even pool size is higher than the paper's threshold, rising with discount rate and falling with the length of the attack.

ECONOMIC MECHANISM. A transition loss in blocks per day is repaid by a gain after retargeting.

LITERATURE CONNECTION. Grunspan and Pérez-Marco (2018); Sapirshtein et al. (2016).

DATA. Public Bitcoin block data (timestamps, difficulty); estimated pool shares from public pool statistics.

VARIABLES. Pool share, γ , cost per hash, price, discount rate, horizon.

METHOD. Markov decision process with retargeting; Monte Carlo for validation.

IDENTIFICATION STRATEGY. Not econometric; the model is solved exactly and calibrated on observed parameters.

EXPECTED CONTRIBUTION. A corrected threshold that depends on time.

MAIN RISK. Results depend on assumed costs and discount rate.

Idea 2. Detecting withholding in historical block data

RESEARCH QUESTION. Is there statistical evidence that some pools deviate from honest mining, as measured by the relation between their share of blocks and the rate and length of forks involving their blocks?

MOTIVATION. The attack has been discussed for years without systematic evidence of its use.

HYPOTHESIS. Under honest mining, orphan and tie outcomes are independent of the pool identity once propagation is controlled for; deviations appear as excess wins in ties.

ECONOMIC MECHANISM. Selfish pools win more ties and orphan more honest blocks.

LITERATURE CONNECTION. Eyal and Sirer (2014); Decker and Wattenhofer (2013).

DATA. Full block history, node-level first-seen times where available, pool attribution by coinbase tag.

VARIABLES. Pool of each block, fork occurrence and length, inter-block times, propagation proxies.

METHOD. Likelihood ratio test of a honest-mining Markov model against Selfish-Mine; logistic regression of tie wins.

IDENTIFICATION STRATEGY. Compare pools with similar connectivity; use exogenous network events as controls.

EXPECTED CONTRIBUTION. First systematic test of whether the attack is used.

MAIN RISK. Low power: ties are rare, and first-seen data may be incomplete.

Idea 3. Market pricing of consensus-security thresholds

RESEARCH QUESTION. Do cryptocurrency returns react when a mining pool crosses a size at which attacks become profitable?

MOTIVATION. The paper's thresholds are testable against asset prices.

HYPOTHESIS. Negative abnormal returns around threshold crossings, larger for lower γ proxies.

ECONOMIC MECHANISM. Investors price the probability of control by a single pool.

LITERATURE CONNECTION. Budish (2018); Biais et al. (2019).

DATA. Daily exchange prices, estimated pool shares, a market index of crypto assets.

VARIABLES. Abnormal returns, pool share, indicator of threshold crossing, volume.

METHOD. Event study with a market model and placebo events.

IDENTIFICATION STRATEGY. Crossings are driven by pool dynamics, plausibly exogenous to price in short windows.

EXPECTED CONTRIBUTION. Links consensus theory to asset pricing.

MAIN RISK. Few events and confounding news.

Idea 4. Pool formation as a dynamic game

RESEARCH QUESTION. What distribution of pool sizes is an equilibrium when miners choose pools to maximise revenue, with switching costs and risk aversion?

MOTIVATION. The paper conjectures a drift to a majority pool without a model.

HYPOTHESIS. Variance reduction and selfish premia favour concentration, but switching costs and the possibility of attack-induced price falls cap it.

ECONOMIC MECHANISM. Miners trade off a variance benefit and a selfish premium against the loss of value from concentration.

LITERATURE CONNECTION. Kiayias et al. (2016); Rosenfeld (2011).

DATA. Time series of pool shares and concentration indices from public sources.

VARIABLES. Pool shares, hash power, fees, price.

METHOD. Evolutionary game or Markov-perfect equilibrium, with calibration to the share dynamics.

IDENTIFICATION STRATEGY. Match the model to the observed concentration path.

EXPECTED CONTRIBUTION. Turns a verbal argument into a testable theory.

MAIN RISK. Weak identification of switching costs.

Idea 5. Selfish strategies under proof of stake

RESEARCH QUESTION. Does an analogue of Selfish-Mine, based on withholding or reordering blocks and attestations, yield a profitable threshold in proof-of-stake protocols?

MOTIVATION. Much of the value in settlement systems now sits on proof-of-stake chains.

HYPOTHESIS. Thresholds differ, because stake cannot be wasted on orphaned work but proposers can be reordered for extractable value.

ECONOMIC MECHANISM. The reward of a proposer depends on fork choice and on the value of ordering transactions.

LITERATURE CONNECTION. Daian et al. (2020); Kiayias et al. (2016).

DATA. Public beacon-chain data for the empirical part; simulation for the theory.

VARIABLES. Stake share, proposal outcomes, reorganisations, fees.

METHOD. A Markov model for the fork choice rule and simulation.

IDENTIFICATION STRATEGY. Model-based; the empirical part compares observed reorganisation rates with model predictions.

EXPECTED CONTRIBUTION. Extends the analysis to the dominant design.

MAIN RISK. Protocol details differ and change quickly.

11.1 Ranking

Table 5. Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Finite-horizon profitability	3	5	5	4	4	4	25
2. Detection in block data	4	3	4	4	4	5	24
3. Market pricing of thresholds	4	4	4	3	4	4	23
4. Pool formation game	3	3	3	4	4	4	21
5. Proof-of-stake analogue	4	3	3	3	5	4	22

Source: reviewer's subjective scoring; no official or empirical basis.

EXTENSION The top three are ideas 1, 2 and 3. Idea 1 is the cheapest and answers the main weakness exactly; idea 2 provides the missing empirical counterpart and would be a contribution in its own right; idea 3 is the one that connects the mechanism to asset pricing and is the most natural project for a finance audience.

12 Learning Framework

12.1 Prerequisite concepts

- Proof of work, hash puzzles, the longest-chain rule, orphaned blocks and difficulty adjustment.
- Poisson processes and the memoryless property of exponential waiting times.
- Discrete-time Markov chains, stationary distributions and balance equations across cuts.
- Geometric series, and the notion of a Nash equilibrium and of incentive compatibility.

12.2 Core concepts learned

- A majority assumption on honesty does not imply incentive compatibility: a minority can profit by withholding.
- The threshold of profitability depends on how ties are resolved, summarised by γ .
- A one-dimensional state (the lead) is enough to analyse a strategic race.

12.3 Important equations

Equations (1), (3), (6) and (7) are the ones worth remembering.

12.4 Methods learned

Cut equations for stationary distributions; reward accounting by event; checking a closed form by an event-by-event simulation; reducing a game to a threshold in a single parameter.

12.5 Questions to test understanding

COMPREHENSION

1. In which state does the pool take risk, and why?
2. What does γ represent, and why does the current protocol let a pool push it towards one?
3. Why does the total rate of blocks in the chain fall under selfish mining?

ANALYSIS

1. Derive p_1 from the balance equations, and explain where $\alpha < 1/2$ is used.
2. Derive the threshold $(1 - \gamma)/(3 - 2\gamma)$ from Equation (6).
3. Why is relative revenue not a measure of profit, and what happens before the difficulty adjusts?

RESEARCH

1. How would you detect selfish mining in block data?
2. How would the result change with several selfish pools?
3. What is the analogue of the attack in a proof-of-stake chain?

12.6 Papers to read next

Sapirshtein, Sompolinsky and Zohar (2016); Kiayias et al. (2016); Grunspan and Pérez-Marco (2018); Biais et al. (2019); Budish (2018).

13 Conclusion

13.1 One-sentence thesis

A minority mining pool can earn more than its share of rewards by withholding blocks, with a profitability threshold between zero and one third that depends on how ties are resolved.

13.2 Three most important insights

- Security against a minority is not the same as incentive compatibility: honesty is not an equilibrium of the mining game.
- The decisive parameter is the tie-breaking share γ , which depends on the network and not on computing power.
- The result is about relative revenue; whether it is profit depends on retargeting, cost and horizon.

13.3 Strongest evidence

The closed-form revenue (Equation 6) and its threshold, which reproduce exactly from the state machine and agree with simulation.

13.4 Biggest weakness

The criterion is relative revenue and not profit: before the difficulty adjusts, the pool earns fewer blocks per unit of time than if it were honest.

13.5 Most important research gap

Empirical evidence on whether and where the strategy has been used, and measurement of γ .

13.6 Best research opportunity

A finite-horizon, retargeting-aware profitability analysis, followed by statistical detection in block data.

13.7 Three papers to read next

Sapirshstein, Sompolinsky and Zohar (2016); Grunspan and Pérez-Marco (2018); Biais et al. (2019).

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